# MCO-1D ORNITHOPTER

Commissioning Manual — Revision 16 — July 2026

The complete guide from first assembly through experimental Discovery flights. Assumes no prior knowledge of this project. Every term is defined on first use. Every step is numbered. Do not skip steps.

| | |
|---|---|
| **Aircraft** | MCO-1D metachronal cascade ornithopter |
| **Mission** | Search and Rescue testbed, biomimetic flight research |
| **MTOW** | 3.448 kg (7.60 lb) — measure actual after build |
| **Wings** | 8 independent — four tandem pairs W1 (front) to W4 (rear) |
| **Stall speed** | 4.85 m/s (9.4 kt) out of ground effect |
| **VNE** | 28.0 m/s (54.4 kt) |
| **Hover freq (est)** | 2.5 Hz — measured precisely at TC-02 |
| **Freq limit** | 3.0 Hz hard cap — cannot be overridden |
| **Motors** | 8 × GoolRC 5010-14 360 KV — one per wing, all independent |
| **ODrives** | 4 × v3.6 — axis0 = port, axis1 = starboard per pair |
| **Pitot airspeed** | Pixhawk MS4525DO → V_ms — used by all morphing hypotheses |
| **Wave Library** | 48,001 lines — 33 hypotheses H-01 to H-36, 180 variants |
| **GS commands** | 148 — type "help" on ground station for full list |
| **Tick rate** | 50 Hz — 20 ms per tick, ~13 ms mean on Pi 4 |
| **Test cards** | TC-01 bench, TC-02 captive hover, TC-03 truck, TC-04 free flight |
| **Revision** | 16 — July 2026 |

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# PART I — WHAT THIS AIRCRAFT IS AND HOW IT WORKS

## SECTION 1 — WHAT THE MCO-1D IS

The MCO-1D is a prototype aircraft that flies by flapping its wings — the same way a bird or large insect does. It has eight wings in four tandem pairs. The front pair is W1, the second pair W2, then W3, then W4 at the rear. Each pair has a port (left) wing and a starboard (right) wing — eight wings total, all completely independent.

Every wing has its own motor. The computer commands all eight motors fifty times per second. This is what makes the MCO-1D different from any previous ornithopter: each of the eight wings can do something different at the same moment, and the computer can change what any wing is doing within 20 milliseconds.

The mission is Search and Rescue. The aircraft is designed to thread through confined spaces, hover in wind that would ground a conventional drone, and operate at low altitude where rotor-wash from a multirotor would be dangerous to people on the ground.

## SECTION 2 — THE EIGHT WINGS AND METACHRONAL CASCADE

The wings do not all flap at the same time. They flap in a staggered sequence called a metachronal cascade: W1 starts, then W2 begins a fraction of a cycle later, then W3, then W4. Each rearward pair captures energy from the air disturbed by the pair ahead. Research on plesiosaurs showed correctly phased hind flippers generated 60% more thrust and 40% better efficiency than working alone. The MCO-1D uses the same principle with four pairs instead of two.

Each wing can independently control five parameters:

| Parameter | What It Controls |
|---|---|
| Amplitude | How far the wing swings in its arc. 0–96°. Hard limit 96°. |
| Frequency | How many flaps per second. 0.5–3.0 Hz. Hover ~2.5 Hz. |
| Angle of Attack (AoA) | Pitch of wing surface to airflow. 0–22°. Best lift ~15–18°. Stall above 22°. |
| Sweep | Fore-aft position of the wing. Also controls morphing stow. |
| Dihedral | Up-down tilt. Used in steep approach braking. Open-loop. |

## SECTION 3 — WHAT THE COMPUTER KNOWS — PER-WING SENSING

The MCO-1D knows the actual aerodynamic state of every individual wing in real time. This is unprecedented in any flapping-wing aircraft.

| What is Known | How |
|---|---|
| Angle of Attack per wing | MCP3208 ADC #1 reads the pitch servo feedback potentiometer voltage. Actual AoA per wing at 0.044° resolution, 50 Hz. |
| Sweep position per wing | MCP3208 ADC #2 reads the sweep servo feedback voltage. Actual fore-aft position per wing, 50 Hz. |
| Stroke position per wing | ODrive pos_estimate via CAN. Actual flap angle in the stroke arc. No new hardware — the ODrive already knows this. |
| Effective AoA | Computed: geometric AoA plus the velocity the wing tip adds as it moves through the air. Tip velocity can exceed 6 m/s at mid-stroke. |

| | |
|---|---|
| Pitot airspeed | Pixhawk MS4525DO pitot tube → msg.airspeed → V_ms. Used by all morphing hypotheses H-33/34/35/36. NOT GPS ground speed — GPS is only for navigation. |

## SECTION 4 — WING MORPHING — THE SPEED AND ENDURANCE PROGRAMME

This is the most important new capability added in Revision 16. Any wing pair can be placed in one of three states independently:

| Wing State | What It Does |
|---|---|
| State A — Active flapping | Normal. Wing oscillates at commanded amplitude and frequency. Generates lift and thrust via unsteady aerodynamics. |
| State B — Stopped at neutral | Wing held stationary at best-glide AoA. Zero motor power. At forward speed V, generates fixed-wing lift: L = ½ρV²S·CL. At 8 m/s, two stopped pairs generate more lift than MTOW — the active pairs become pure thrust engines. |
| State C — Stowed along fuselage | Sweep servo drives wing to maximum-aft position (~85-95°). Wing lies approximately parallel to fuselage. Effective wing area → near zero. Gust loading → near zero. Active pairs must carry full MTOW lift. |

Four hypotheses implement the morphing programme:

| Hyp | Name (variants) | Purpose |
|---|---|---|
| H-33 | Variable wing area (18v) | Stow and stop pairs. Trim via active-pair sweep. |
| H-34 | Dynamic sweep kinematics (16v) | ±20-40° sweep within each stroke. Upstroke aft cuts drag 69%. |
| H-35 | Thrust/lift separation (14v) | Stopped pairs = wings. Active pairs = engines. Pure role split. |
| H-36 | Loiter and ferry (8v) | 3 pairs stopped as wings. 1 pair at minimum freq. ~3× endurance. |

✓ All morphing decisions use pitot airspeed from the Pixhawk pitot tube. GPS ground speed is never used for any aerodynamic calculation.

## SECTION 5 — SAFETY RULES

■ WARNING: Frequency ceiling 3.0 Hz, amplitude ceiling 96°, and AoA stall protection cannot be overridden. Hard limits coded into the software.

■ WARNING: TC-01 must fully pass before TC-02. TC-02 before TC-03. TC-03 before TC-04. Never add two unknowns at the same time.

■ WARNING: H-33 State C and H-35/36 two-pair configurations are gated on the TC-03 CT-J curve. Do not attempt them before TC-03 completes.

✓ H-33/34/35/36 State B variants (stopped at neutral) are safe for TC-04. Stopped wings still generate fixed-wing lift — the aircraft is not at risk.

✓ The optimizer suggests. The pilot decides everything.

# PART II — HARDWARE: EVERY COMPONENT

## SECTION 6 — WING ASSEMBLIES

| | |
|---|---|
| **Motor model** | GoolRC 5010-14, 360 KV, 7 pole pairs (12N14P) |
| **Pole pairs** | 7 — enter this exact number in the ODrive setup |
| **Gear ratio** | 7:1 — sector wheel turns once per 7 motor turns |
| **Sector wheel arc** | 100° mechanical limit. Software limit: 96° |
| **Rope drive** | Independent per wing. No shared rope between port and stbd on any pair. |
| **Rope inspection** | Before every flight. Fraying shows as rising motor current first. |

## SECTION 7 — ODRIVE MOTOR CONTROLLERS

Four ODrive v3.6 boards, one per pair, two axes each. The assignment is fixed and must match the wiring exactly:

| Board | CAN Node | axis0 | axis1 |
|---|---|---|---|
| Board 0 | Node 0 | axis0 = W1 port | axis1 = W1 starboard |
| Board 1 | Node 1 | axis0 = W2 port | axis1 = W2 starboard |
| Board 2 | Node 2 | axis0 = W3 port | axis1 = W3 starboard |
| Board 3 | Node 3 | axis0 = W4 port | axis1 = W4 starboard |

● CAUTION: vel_limit must be set to 150 rad/s on BOTH axes of ALL FOUR boards (8 axes total). Run mco_odrive_setup.py. Factory default is too low — wings will not reach operating frequency.

## SECTION 8 — PITCH SERVOS AND ANGLE OF ATTACK

Each wing has one pitch servo controlling AoA (0–22°). The servo MUST have a feedback potentiometer output pin — a small voltage proportional to actual servo position, separate from the command signal.

| | |
|---|---|
| **GPIO PWM outputs** | W1p=18, W1s=23, W2p=24, W2s=25, W3p=12, W3s=16, W4p=20, W4s=21 |
| **PWM frequency** | 333 Hz (high-speed digital mode) |
| **Supply** | 7.4 V from dedicated servo LiPo |
| **Feedback wiring** | Feedback pin → 1 kΩ resistor → MCP3208 #1 (SPI1 CE1) |
| **Channel map** | W1p→CH0, W1s→CH1, W2p→CH2, W2s→CH3, W3p→CH4, W3s→CH5, W4p→CH6, W4s→CH7 |

## SECTION 9 — SWEEP SERVOS: FORE-AFT, DYNAMIC SWEEP, AND STOW

Each wing also has one sweep servo controlling the wing fore-aft position. This servo has three distinct functions in the morphing programme:

| Function | Description |
|---|---|
| Normal flight | Holds a fixed position. 60° neutral. |

| | |
|---|---|
| H-34 dynamic sweep | Moves ±20-40° within each stroke — forward on downstroke, aft on upstroke. Must execute within 200 ms. |
| H-33 State C stow | Drives to ~85-95° aft — wing lies along fuselage. Measure the actual achievable stow angle with mco_build_measurements.py. |
| H-33/35 trim compensation | When a pair is stowed, other pairs adjust sweep to re-trim pitch moment. No fixed pitch trim penalty from any stow configuration. |

| | |
|---|---|
| **GPIO PWM outputs** | W1p=4, W1s=17, W2p=27, W2s=22, W3p=5, W3s=6, W4p=13, W4s=19 |
| **Feedback wiring** | Feedback pin → 1 kΩ resistor → MCP3208 #2 (SPI1 CE0) |
| **Stow requirement** | Sweep servo must reach 85-95° aft for H-33 State C. Measure actual range before TC-01. |

## SECTION 10 — ADC CHIPS

| | |
|---|---|
| **Both chips share** | SPI1: CLK=GPIO11, MOSI=GPIO10, MISO=GPIO9 |
| **Pitch ADC (MCP3208 #1)** | SPI1 CE1 = GPIO7 |
| **Sweep ADC (MCP3208 #2)** | SPI1 CE0 = GPIO8 |
| **Power** | VDD and VREF both to Pi 3.3 V |
| **Resolution** | 12-bit → 0.8 mV/step → ~0.044°/step |
| **/boot/config.txt** | dtoverlay=spi1-2cs |
| **Series resistors** | 1 kΩ on each of 16 signal lines |

## SECTION 11 — PITOT TUBE — AIRSPEED FOR THE MORPHING PROGRAMME

■ WARNING: The pitot tube is not optional for H-33/34/35/36. Every morphing decision in those four hypotheses uses pitot airspeed. GPS ground speed is never used for any aerodynamic calculation.

The Pixhawk accepts a standard MS4525DO digital airspeed sensor on its I2C airspeed port. The sensor measures the pressure difference between the stagnation port (facing forward) and the static port (side holes). The Pixhawk converts this to airspeed and sends it via MAVLink VFR_HUD messages. mco_mavlink.py reads msg.airspeed and puts it into the sensors dict as V_ms. Every tick, every morphing hypothesis reads this value. The label h34_V_ms_used and h35_V_ms_pitot in the result dict confirm that pitot airspeed was used.

| | |
|---|---|
| **Sensor** | Pixhawk-compatible MS4525DO digital airspeed |
| **Mount** | Forward fuselage nose or forward boom — clean undisturbed air |
| **Connection** | Pixhawk airspeed port (I2C) |
| **Software path** | Pixhawk → MAVLink VFR_HUD → mco_mavlink.py → sensors[V_ms] |
| **Used by** | H-33 speed-adaptive stow trigger, H-34 stopped-wing sweep schedule, H-35 role switching thresholds, H-36 free lift calculations, morphing_speed_score() composite scorer |

## SECTION 12 — LOAD CELL — COMMISSIONING BENCH INSTRUMENT

The load cell is used only during TC-01 and TC-02. It is not flown. It sits inline between the aircraft and the static test post. As wing frequency increases, the load cell reads less force (aircraft is generating more lift). When the load cell reads zero, the aircraft is generating lift exactly equal to its own weight — that is the hover frequency. TC-02 is entirely built around this measurement.

| | |
|---|---|
| **Capacity** | 50 N minimum. 100 N preferred for headroom. |
| **Type** | Compression load cell — aircraft pushes down, cell reads force. |
| **ADC** | HX711 — DT=GPIO26, SCK=GPIO7 |
| **Software** | mco_loadcell.py — calibrate before TC-01 with known 1 kg mass |
| **Calibration** | Run: python3 mco_loadcell.py --calibrate |
| **TC-01 use** | Thrust curve — force vs frequency at each step |
| **TC-02 use** | Hover detection — reads zero when lift = weight |
| **Hover frequency** | Frequency at which load cell first reads ≤1 N |
| **Pin conflict note** | GPIO7 is also SPI1 CE1 (pitch ADC). No conflict: HX711 is bench-only; SPI1 and HX711 not active simultaneously. |

## SECTIONS 13–17 — Pi 4, Sensors, Power, Radio, Panel

These sections are unchanged from Revision 15. See the GPIO reference in Section 18 and the BOM for full specifications.

| | |
|---|---|
| **Pi 4** | 4 GB RAM. OS Lite 64-bit. 64 GB microSD high-endurance. |
| **IMU** | MPU-6050 at I2C 0x68. 100 Hz. Vibration-isolate mount. |

| | |
|---|---|
| **Barometer** | BMP280 at I2C 0x76. Same I2C1 bus as IMU. |
| **Lidar** | TF-Luna, 50 Hz, facing down, clear of wing arc. |
| **OAT** | DS18B20, GPIO4, One-Wire, 4.7 kΩ pull-up. |
| **Main battery** | 4S LiPo 4,000 mAh+. Battery interlock: 80% min for TC ops. |
| **Servo battery** | 2S LiPo 1,500 mAh. Dedicated 7.4 V for all 16 servos. |
| **Pi supply** | Isolated BEC 5V/3A. Never share with motor supply. |
| **RC** | SBUS receiver. CH1=Roll, CH2=Pitch, CH3=Lever, CH4=Yaw. |
| **Pixhawk** | Sensor bridge only. GPS, compass, IMU, pitot → Pi via MAVLink. |
| **Panel** | mco_panel.py — 4 buttons, 6 margin bars (amber 50%, red 25%). |

## SECTION 18 — WIRING REFERENCE

| Pin | Function |
|---|---|
| GPIO 4 | DS18B20 OAT (One-Wire) |
| GPIO 5 | W3p sweep servo PWM |
| GPIO 6 | W3s sweep servo PWM |
| GPIO 7 | SPI1 CE1 — MCP3208 #1 pitch ADC chip select / HX711 SCK (bench) |
| GPIO 8 | SPI1 CE0 — MCP3208 #2 sweep ADC chip select |
| GPIO 9 | SPI1 MISO — both ADC chips |
| GPIO 10 | SPI1 MOSI — both ADC chips |
| GPIO 11 | SPI1 CLK — both ADC chips |
| GPIO 12 | W3p pitch servo PWM |
| GPIO 13 | W4p sweep servo PWM |
| GPIO 16 | W3s pitch servo PWM |
| GPIO 17 | W1s sweep servo PWM |
| GPIO 18 | W1p pitch servo PWM |
| GPIO 19 | W4s sweep servo PWM |
| GPIO 20 | W4p pitch servo PWM |
| GPIO 21 | W4s pitch servo PWM |
| GPIO 22 | W2s sweep servo PWM |
| GPIO 23 | W1s pitch servo PWM |
| GPIO 24 | W2p pitch servo PWM |
| GPIO 25 | W2s pitch servo PWM / CAN adapter interrupt |
| GPIO 26 | HX711 DT — load cell data (bench use only) |
| GPIO 27 | W2p sweep servo PWM |
| I2C1 GPIO2/3 | IMU (0x68) + barometer (0x76) |
| SPI0 CE0 | MCP2515/2518FD CAN bus adapter |
| UART GPIO14/15 | RC receiver SBUS + Pixhawk MAVLink at 921,600 baud |

# PART III — ASSEMBLY

## SECTION 19 — ASSEMBLY STEPS 1 THROUGH 24

■ WARNING: Do not connect any battery until Step 18. Do not power ODrives until mco_odrive_setup.py has been run (Step 25).

| Step | Action | Expected / Notes |
|---|---|---|
| 1 | Mount airframe | All structural members assembled. No flex between motor mounts. |
| 2 | Mount 8 sector wheels | One per wing. Rotate by hand. Full arc smooth, no binding. |
| 3 | Mount 8 motors | Thread lock on all motor mount screws. Verify shaft alignment. |
| 4 | Install 8 rope drives | One per wing, independent. Verify 7:1 ratio: 7 motor turns = 1 sector turn. |
| 5 | Mount 8 wings | Attach to sector wheel pivot points. Swing through full 96° without binding. |
| 6 | Set pitch servo neutrals | Apply servo power only. Command all pitch servos to 0°. Verify mechanical neutral. Attach horn at that position. |
| 7 | Install 8 pitch servos | One per wing. Thread feedback wire back along servo cable. |
| 8 | Set sweep servo neutrals | Command all sweep servos to 60° (neutral). Attach horn. Then command to 95° and measure actual angle reached (for H-33 stow). |
| 9 | Install 8 sweep servos | One per wing. Verify linkage does not bind at max-aft position. |
| 10 | Mount 4 ODrive boards | Vibration isolation mounts. Wire motor phases: Board N axis0 = W(N+1) port, axis1 = W(N+1) stbd. |
| 11 | Wire CAN bus | Pi adapter → Board 0 → 1 → 2 → 3. Twisted pair. 120 Ω terminator at Pi end AND at Board 3 end. |
| 12 | Wire pitch servo PWM | Pi GPIO to pitch servo signal wires per Section 18. |
| 13 | Wire sweep servo PWM | Pi GPIO to sweep servo signal wires per Section 18. |
| 14 | Wire pitch servo feedback | Each pitch feedback pin → 1 kΩ → MCP3208 #1 (SPI1 CE1) channel. |
| 15 | Wire sweep servo feedback | Each sweep feedback pin → 1 kΩ → MCP3208 #2 (SPI1 CE0) channel. |
| 16 | Mount Pi 4 and sensors | Pi on vibration isolation. IMU and baro to I2C1 (GPIO2 SDA, GPIO3 SCL). Lidar facing straight down, clear of wing arc. |
| 17 | Connect OAT and pitot sensors | DS18B20 data → GPIO4, 4.7 kΩ pull-up to 3.3 V. Pitot tube → Pixhawk airspeed port. Mount pitot tube facing forward in clean airflow. |

| | | |
|---|---|---|
| 18 | Route power cables — DO NOT CONNECT BATTERIES YET | Main battery → 30A fuse → ODrives. Servo battery → all servo rails. Isolated BEC → Pi. |
| 19 | Install RC receiver | SBUS output → Pi UART GPIO15 RXD. Bind to transmitter now. |
| 20 | Connect Pixhawk | Pixhawk TELEM2 → Pi UART at 921,600 baud. |
| 21 | First power-on — Pi only (no motors, no servos) | Connect isolated BEC to Pi only. Pi boots. SSH in. python3 --version → should show 3.8 or later. |
| 22 | Install software | See Part IV. Complete all 9 software steps before Step 23. |
| 23 | Run ODrive configuration | python3 mco_odrive_setup.py — configures all 8 axes. ~2 minutes. |
| 24 | Run build measurements | python3 mco_build_measurements.py — measure 15 physical dimensions. Patches wave library with real values. Required before TC-01. |

## PART IV — SOFTWARE INSTALLATION

Nine steps. Do them in order. $ means type this into a terminal. Do not type the $ itself.

### STEP 20 — INSTALL RASPBERRY PI OS

On your laptop: download Raspberry Pi Imager from raspberrypi.com/software. Insert 64 GB microSD. Open Imager:

| Setting | Value |
|---|---|
| Device | Raspberry Pi 4 |
| OS | Raspberry Pi OS Lite (64-bit) — not the desktop version |
| Storage | Your 64 GB microSD card |

Click gear (Advanced Options): hostname=mco-1d, enable SSH, set username/password, configure WiFi, set locale. Click Write.

### STEP 21 — CONFIGURE /boot/config.txt

Before first boot: open the SD card on your laptop. Edit config.txt. Add at the end:

```
# CAN bus adapter
dtoverlay=mcp2515-can0,oscillator=8000000,interrupt=25

# Both MCP3208 ADC chips on SPI Bus 1
dtoverlay=spi1-2cs

# OAT sensor on GPIO4
dtoverlay=w1-gpio,gpiopin=4

# Enable I2C and SPI
dtparam=i2c_arm=on
dtparam=spi=on
```

### STEP 22 — INSTALL PYTHON PACKAGES

Boot Pi. SSH in: $ ssh pi@mco-1d.local

```
$ pip3 install pymavlink python-can pigpio spidev smbus --break-system-packages
$ sudo systemctl enable pigpiod && sudo systemctl start pigpiod
```

✓ --break-system-packages is required on Raspberry Pi OS Bookworm and later.

### STEP 23 — COPY SOFTWARE FILES

From your laptop (not from inside the SSH session):

```
$ scp mco_wave_library.py mco_main.py mco_odrive_interface.py \
mco_loadcell.py mco_build_measurements.py \
mco_preflight_checks.py mco_rc_verify.py \
mco_vibration_check.py mco_test_cards.py mco_log_analysis.py \
mco_servo_interface.py mco_mavlink.py mco_gs_server.py \
```

```
mco_logger.py mco_odrive_setup.py mco_preflight.py \
mco_panel.py mco_hypothesis_tool.py mco_sync.py \
mco_ground_station.py mco_can_setup.sh mco_install.sh \
pi@mco-1d.local:/home/pi/
```

```
$ ssh pi@mco-1d.local
$ mkdir -p /home/pi/logs /home/pi/knowledge
$ chmod +x /home/pi/mco_can_setup.sh /home/pi/mco_install.sh
$ bash /home/pi/mco_install.sh
```

## STEP 24 — CONFIGURE THE CAN BUS

```
$ sudo reboot
$ ssh pi@mco-1d.local
$ sudo bash /home/pi/mco_can_setup.sh
$ ip link show can0 # must show UP
```

## STEP 25 — CONFIGURE ODRIVES (ALL 8 AXES)

Apply main battery power. Then:

```
$ python3 /home/pi/mco_odrive_setup.py
```

This configures all 4 boards, both axes each: pole_pairs=7, vel_limit=150 rad/s, MOTOR_TYPE_HIGH_CURRENT, velocity control mode. Saves to non-volatile memory. ~2 minutes.

● CAUTION: If any board does not respond: check CAN bus wiring and both 120 Ω termination resistors. Run: candump can0 to see traffic.

## STEP 26 — PRE-FIRST-FLIGHT SEQUENCE

Run these tools in order before TC-01:

| Step | Action | Expected / Notes |
|---|---|---|
| 26a | Load cell calibration | python3 mco_loadcell.py --calibrate Use a 1.000 kg calibration mass. Saves to /home/pi/knowledge/loadcell_cal.json. |
| 26b | Build measurements | python3 mco_build_measurements.py Measure all 15 dimensions. ~30 minutes. Patches wave library with real values. |
| 26c | RC channel verification | python3 mco_rc_verify.py Confirm CH1=Roll, CH2=Pitch, CH3=Lever, CH4=Yaw. Verify lever direction. Saves to /home/pi/knowledge/rc_mapping.json. |
| 26d | Vibration baseline | python3 mco_vibration_check.py --quick IMU baseline before any motors run. Quick mode: no motor sweep. |

## STEP 27 — VERIFY ALL SYSTEMS

Run the preflight check and verify all 14 checks pass:

```
$ python3 mco_preflight_checks.py --bench
```

All 14 checks must return PASS or WARN. Any FAIL must be resolved before proceeding.

| Check | What it verifies |
|---|---|
| Wave library | Import success, correct line count |
| Build measurements | Confirmed from mco_build_measurements.py |
| SD card space | At least 500 MB free |
| pigpiod | Active — servo PWM daemon running |
| CAN bus | can0 UP |
| ODrives | All 8 axes responding |
| ODrive vel_limit | 150 rad/s confirmed on at least axis 0 |
| Pitch ADC | All 8 channels readable |
| Sweep ADC | SPI1 CE0 readable |
| IMU | 0x68 responding, az within 0.1g of 1.0g |
| Barometer | 0x76 responding |
| OAT sensor | DS18B20 reading plausible temperature |
| Battery voltage | Main ≥80% for TC ops (arming interlock) |
| Load cell | HX711 responding, calibration file loaded |

## STEP 28 — VIBRATION FULL SWEEP

After TC-01-06 (motor spin test) passes, run the full vibration sweep before TC-01-07 (first wing flapping):

```
$ python3 mco_vibration_check.py
```

This sweeps 0.5–3.0 Hz, measures IMU RMS and SPI reliability at each frequency, and flags if vibration damping is inadequate.

## STEP 29 — GROUND STATION LAPTOP SETUP

On your laptop:

```
$ python3 mco_ground_station.py mco-1d.local 5760
```

Shows live telemetry at 10 Hz. Type "help" for all 148 commands. For the per-wing sensing and morphing display:

```
>>> sensing_display
>>> morphing_speed_report
>>> h36_status
```

## PART V — COMMISSIONING TESTS

Four test cards, each building on the last. Never advance to the next card until the current one fully passes. Never add two unknowns at the same time.

| Card | Type | Physical Arrangement |
|---|---|---|
| TC-01 | Bench test | Aircraft bolted to static post with load cell. No movement. 16 steps. 10 fully automatic, 6 semi-automatic. |
| TC-02 | Captive hover | Same post, same load cell. Fully automatic. Steps frequency up until load cell reads zero. That IS the hover frequency. |
| TC-03 | Truck-mount forward flight | Aircraft on truck bed. Truck provides airspeed. Fully automatic. Maps CT-J curve and forward flight aerodynamics. |
| TC-04 | First free flight | Unrestrained. Pilot in command. Software guides step by step. 7 steps. |

### SECTION 29 — TC-01 BENCH TEST: 16 STEPS

All tests with aircraft bolted to post, load cell inline, no movement. Start from ground station: tc01_start Monitor progress: tc01_status Confirm physical actions: tc01_confirm

| Step | Action | Expected / Notes |
|---|---|---|
| TC-01-01 | Servo neutral check — AUTO | Software commands all 16 servos to neutral. Reads MCP3208 actual positions. PASS: all within ±2°. |
| TC-01-02 | Servo range ±45° — AUTO | Commands ±45°. Reads ADC at each position. PASS: all reach commanded range within ±3°. |
| TC-01-03 | Pitch ADC all 8 channels — AUTO | Reads MCP3208 #1 five times. PASS: all 8 channels return non-None values. |
| TC-01-04 | Sweep ADC all 8 channels — AUTO | Reads MCP3208 #2 via SPI1 CE0. PASS: all 8 channels readable. |
| TC-01-05 | ODrive 8-axis comms — AUTO | Reads encoder estimates from all 4 boards, both axes each. PASS: all 8 respond. |
| TC-01-06 | Motor spin at 10 rad/s — AUTO | All 8 axes at 10 rad/s for 2 seconds. Reads currents. PASS: all currents 0.3–6.0 A. Run mco_vibration_check.py after this step. |
| TC-01-07 | Wing flapping at 0.5 Hz — AUTO | 30 seconds at 0.5 Hz. Monitors rope via position read. PASS: no rope slip detected, all wings tracking. |
| TC-01-08 | Wing flapping at 2.5 Hz — AUTO | 60 seconds at 2.5 Hz. Monitors motor temperature via current proxy. Verify with IR thermometer: all motors below 60°C. PASS: sustained 60 s without thermal limit. |

| | | |
|---|---|---|
| TC-01-09 | Pitch ADC calibration — SEMI-AUTO | Software commands each pitch servo to -45°, 0°, +45° in turn. PAUSES and asks operator to visually confirm wing is at commanded position. Type tc01_confirm at each position. Software records ADC voltage and computes calibration. PASS: all 8 pitch servos calibrated, round-trip error <1°. |
| TC-01-10 | Sweep ADC calibration — SEMI-AUTO | Same procedure as TC-01-09 for sweep servos. PASS: all 8 sweep servos calibrated. |
| TC-01-11 | Frequency ceiling test — AUTO | Steps 2.0 to 3.0 Hz in 0.1 Hz steps, 30 s each. Monitors temperature proxy. Records thermal ceiling. PASS: ceiling at or above 2.5 Hz. |
| TC-01-12 | Static thrust measurement — SEMI-AUTO | Operator confirms aircraft secured to load cell fixture. tc01_confirm. Software commands 1.5 to 3.0 Hz, reads HX711 at each step. Builds thrust-vs-frequency curve. Saves to /home/pi/knowledge/. PASS: maximum thrust ≥ MTOW (33.8 N) at some frequency ≤3.0 Hz. |
| TC-01-13 | RC channel mapping — SEMI-AUTO | Software asks operator to move each stick. Reads RC channels, identifies which channel moved. PASS: Roll=CH1, Pitch=CH2, Lever=CH3, Yaw=CH4. |
| TC-01-14 | Safety caps and fault detection — SEMI-AUTO | Automated: verify 3.0 Hz cap fires when 3.5 Hz commanded. Automated: verify 96° cap fires when 105° commanded. Semi-auto: operator disconnects W1p motor, confirms fault detection fires, reconnects. PASS: both caps active, fault detected within 0.2 s. |
| TC-01-15 | Port/starboard symmetry — AUTO | Commands all wings to 15° AoA. Reads actual AoA via MCP3208. Computes port vs stbd lift estimate. PASS: asymmetry below 5%. |
| TC-01-16 | Compile results and export — AUTO | Compiles all results into JSON. Exports knowledge to USB. Run: python3 mco_log_analysis.py --tc01 to review thrust curve. |

✓ After TC-01-06: run mco_vibration_check.py before proceeding to TC-01-07. Vibration damping must be confirmed adequate before first wing flapping.

## SECTION 30 — TC-02 CAPTIVE HOVER: 8 STEPS

Same post as TC-01. Same load cell inline. Aircraft still bolted. No pilot. No free flight. Fully automatic. Start: tc02_start Monitor: tc02_status

| Step | Action | Expected / Notes |
|---|---|---|
| TC-02-01 | Load cell check — AUTO | Reads HX711. Verifies reading is plausible relative to MTOW. PASS: load cell responsive and calibration file loaded. |
| TC-02-02 | Lift-frequency curve — AUTO | Steps 1.5 to 3.0 Hz. At each step, reads load cell for 5 seconds. Computes lift = baseline_force - current_force. Fits slope dL/df (lift per Hz). This curve calibrates the descent controller. |
| TC-02-03 | Find hover frequency — AUTO | Steps frequency up from 1.0 Hz in 0.1 Hz increments. At each step, reads load cell. HOVER FREQUENCY = frequency at which load cell reads ≤1 N. This is the most important measurement in the programme. Records hover_freq_hz to knowledge file. |
| TC-02-04 | Confirm hover — AUTO | Holds at hover frequency for 5 seconds. Verifies load cell stays near zero. PASS: mean residual force ≤2 N over 5 seconds. |
| TC-02-05 | AoA sweep at hover frequency — AUTO | Holds at hover frequency. Commands AoA 8°, 12°, 15°, 18°, 20°, 22° in turn. Reads load cell lift at each AoA. Builds CL-alpha curve. Identifies best-lift AoA in actual operating conditions. |
| TC-02-06 | H-26 side force measurement — AUTO | Engages 4° differential AoA (port vs stbd) while restrained. Reads IMU roll rate and lateral acceleration. Classifies: ROLL (roll rate >2°/s) or LATERAL_FORCE (lateral g >0.02g) or UNCLEAR. This is the most uncertain prediction in the programme — only measurement reveals it. |
| TC-02-07 | Calibrate all controllers — AUTO | Applies measured hover_freq_hz and dL/df slope to descent controller. Calls calibrate_descent_controller(). Saves calibration constants to knowledge file. PASS: descent controller accepts calibration. |
| TC-02-08 | 5-minute captive discovery — AUTO | Runs Discovery Mode with aircraft on post. Optimizer explores hover configurations. Exports knowledge at end. Run: python3 mco_log_analysis.py --tc02 to review. |

| | |
|---|---|
| **Key output 1** | hover_freq_hz — actual hover frequency at MTOW |
| **Key output 2** | lift_slope_N_per_Hz — for descent controller calibration |
| **Key output 3** | H-26 classification — ROLL or LATERAL_FORCE |
| **Key output 4** | AoA-CL curve — best-lift AoA in actual operating conditions |

## SECTION 31 — TC-03 TRUCK-MOUNT FORWARD FLIGHT: 6 STEPS

Aircraft mounted on truck bed. Truck provides forward airspeed. Aircraft remains attached to truck (restrained or tethered). No pilot required. Fully automatic. Start: tc03_start Monitor: tc03_status Ground station shows required truck speed. Driver takes it to that speed and holds.

■ WARNING: Pitot tube must be mounted and confirmed reading before TC-03. V_ms must show valid pitot airspeed on ground station telemetry at speed.

| Step | Action | Expected / Notes |
|---|---|---|
| TC-03-01 | Stationary baseline — AUTO | Truck stationary. Commands hover frequency. Reads baseline. |
| TC-03-02 | Transition sweep — AUTO | Ground station shows: "TRUCK: drive at 3.0 m/s". Software waits for pitot airspeed to reach target. Steps frequency downward to find minimum frequency for level flight at each speed. |
| TC-03-03 | CT-J curve — AUTO | At 6 truck speeds from 3–13 m/s, records: pitot airspeed V, wing frequency, computed CT, lift from load cell. Computes advance ratio J = V/(f × 2A × b_mean). This CT-J curve is the key aerodynamic characterisation. Required before H-33 State C, H-35 2-pair, H-36 stow variants. |
| TC-03-04 | Speed envelope — AUTO | At increasing truck speeds, measures IMU stability. Identifies speed at which control authority degrades. PASS: sigma_roll below 5° to at least 11 m/s. |
| TC-03-05 | Descent simulation — AUTO | At cruise speed, steps frequency below hover. Load cell shows whether aircraft has tendency to ascend or descend. Validates descent controller at speed. |
| TC-03-06 | 30-minute discovery at cruise speed — AUTO | Runs morphing_speed discovery mode. Explores H-33/34/35 State B variants at cruise speed. Exports speed leaderboard: morphing_speed_report |

✓ After TC-03: H-33 State C variants, H-35 two-pair separation, and H-36 loiter/ferry are all unlocked. The CT-J curve from TC-03-03 tells you exactly which configurations the remaining active pairs can sustain at each speed.

## SECTION 32 — TC-04 FIRST FREE FLIGHT: 7 STEPS

First unrestrained flight. Pilot in command. Software guides each step and measures automatically. Pilot types tc04_continue to advance. Type tc04_abort at any time — pilot retains full control. Start: tc04_start Monitor: tc04_status

■ WARNING: TC-04 is only attempted when TC-01, TC-02, and TC-03 all pass. Software checks this and blocks tc04_start if any predecessor is incomplete.

| Step | Action | Expected / Notes |
|---|---|---|
| TC-04-01 | Pre-flight confirmation | Software confirms all TC-02/03 calibrations are applied. Pilot verifies site clear, batteries full, RC signal confirmed. tc04_continue to proceed. |
| TC-04-02 | First hover attempts | Ground station instructs: advance lever slowly toward calibrated hover position. Software detects lift-off from rangefinder step change. Records actual free-flight hover frequency. Compare to TC-02 captive result. tc04_continue after 3 clean liftoffs. |
| TC-04-03 | Hover stability 30 seconds | Hold steady hover without correction. Software measures sigma_h. PASS: below 0.5 m. tc04_continue when 30-second hold is complete. |
| TC-04-04 | Altitude hold — descent controller | Command: descent_ctrl_engage 100 (100 fpm). Descend. Verify smooth 100 fpm. Test 300 fpm. Compare to TC-02 captive calibration. tc04_continue. |
| TC-04-05 | Roll response in free flight | Apply 20%, 40%, 60% roll stick. Software records roll rates and settle times. Compare to TC-02 captive estimates. Updates active damping gains from free-flight measurements. |
| TC-04-06 | Forward flight transition | From stable hover: advance lever toward 70%. Allow pitch forward to develop. Fly 100 m forward. Return. Repeat 3 times. Software records transition speed from pitot airspeed. |
| TC-04-07 | First free-flight discovery — 5 minutes | Ground station engages Discovery Mode. Optimizer explores H-33/34/35 State B variants (safe: stopped wings still generate lift). Exports knowledge and speed leaderboard. Run: python3 mco_log_analysis.py after session. |

# PART VI — FLIGHT OPERATIONS

## SECTION 33 — PRE-FLIGHT CHECKLIST

| Step | Action | Expected / Notes |
|---|---|---|
| 1 | python3 mco_preflight_checks.py --flight | All 14 checks must pass. Battery interlock: below 80% = NO-GO for TC. |
| 2 | Area clear | 30 m clear zone. Landing zone identified. No overhead obstacles. |
| 3 | Wind | Below 15 kt normal ops. Below 5 kt for TC-04 first flight. |
| 4 | SD space | ls /home/pi/logs/ — at least 500 MB free. |
| 5 | RC signal | Ground station shows RC VALID. Lever at minimum (safe for arming). |
| 6 | ODrives | Ground station shows all 4 boards, 8 axes IDLE. |
| 7 | Servo health | servo_health_report — all 8 wings Healthy. |
| 8 | Pitot | pitot_airspeed_status — V_ms reading plausibly near zero with aircraft stationary. |
| 9 | ARM | Press ARM. Lever interlock: must be at minimum. Verify correct response directions with small stick inputs. |

## SECTION 34 — NORMAL FLIGHT

| | |
|---|---|
| **Takeoff** | Advance lever smoothly to calibrated hover position from TC-02/04. Smooth input. Aircraft lifts off. Hold hover. |
| **Hover** | Lever at hover position. Active damping and trim balance run automatically. Monitor sigma_h on ground station. |
| **Forward flight** | Advance lever 65-70%. Allow pitch forward. descent_ctrl_engage 0 for altitude hold at speed. |
| **Landing** | Reduce lever for descent. descent_ctrl_engage 100 for 100 fpm precision. Do not cut lever abruptly. |

## SECTION 35 — EMERGENCY PROCEDURES

■ WARNING: LOSS OF CONTROL: Reduce lever to 50%. If aircraft continues unexpected behaviour: full lever aft. Disarm if uncontrollable. Controlled descent to hard landing is better than endangering people.

| | |
|---|---|
| **Wing failure in flight** | CONTINUE (7 wings): land when convenient. LAND AS SOON AS PRACTICABLE (5-6 wings): within 2 min. LAND IMMEDIATELY (4 or fewer): put it down now. |
| **Battery amber (30%)** | Land at first opportunity. |
| **Battery red (20%)** | Land now. Discovery Mode exits automatically. |
| **RC signal loss** | Aircraft holds last config 2 s then descends at 300 fpm. Restore RC. |

| | |
|---|---|
| **Morphing mode anomaly** | h33_h34_h35_disengage_all — returns all pairs to normal flapping. |

## SECTION 36 — POST-FLIGHT

| Step | Action | Expected / Notes |
|---|---|---|
| 1 | DISARM | Press DISARM. All motors stop. |
| 2 | POST-FLIGHT button | 8-minute cooling period. |
| 3 | Export | knowledge_export_usb and morphing_speed_report |
| 4 | Log retrieval | python3 mco_log_analysis.py on laptop after scp of logs. |
| 5 | Battery check | Inspect for puffing or heat. |
| 6 | Rope drives | Inspect all 8. Any fraying = replace before next flight. |
| 7 | Servo health | servo_health_report — flag any wing with rising tracking RMS. |

# PART VII — WING MORPHING PROGRAMME — H-33 TO H-36

## SECTION 37 — OVERVIEW AND AIRSPEED REQUIREMENT

■ WARNING: Every aerodynamic decision in H-33/34/35/36 uses PITOT AIRSPEED from the Pixhawk pitot tube (V_ms = msg.airspeed from VFR_HUD). GPS ground speed is never used for any aerodynamic calculation. The distinction matters because wind can make ground speed and airspeed very different. Only the wing feels airspeed.

Any wing pair can independently be:

| State | Name | Key Property |
|---|---|---|
| State A | Flapping | Normal. Lift + thrust from unsteady aerodynamics. |
| State B | Stopped at neutral | Fixed-wing lift = ½ρV²S·CL. Zero motor power. At 8 m/s pitot, two stopped pairs cover MTOW entirely. |
| State C | Stowed along fuselage | Sweep to ~85-95°. Near-zero wing area. Near-zero gust loading. Full load on active pairs. |

The sweep servos on ACTIVE pairs absorb all pitch trim changes from any stow configuration. There is no fixed pitch penalty from stowing any pair. The optimizer adjusts active-pair sweep continuously.

Free lift from stopped pairs at pitot airspeed (AoA 15°, CL=1.4):

| Pitot airspeed | 1 pair stopped | 2 pairs stopped | 3 pairs stopped |
|---|---|---|---|
| 5 m/s | 1 pair stopped: 7.7 N (23%) | 2 pairs stopped: 15.3 N (45%) | 3 pairs stopped: 23.0 N (68%) |
| 7 m/s | 1 pair stopped: 15.1 N (45%) | 2 pairs stopped: 30.2 N (89%) | 3 pairs stopped: 26.8 N (79%) |
| 8 m/s | 1 pair stopped: 19.7 N (58%) | 2 pairs stopped: 26.9 N (80%) | 3 pairs stopped: 35.0 N (104%) ← |
| 9 m/s | 1 pair stopped: 24.9 N (74%) | 2 pairs stopped: 34.0 N (101%) ← | 3 pairs stopped: 44.3 N (131%) |
| 10 m/s | 1 pair stopped: 30.7 N (91%) | 2 pairs stopped: 42.0 N (124%) ← | 3 pairs stopped: 52.0 N (154%) |

← marks where stopped pairs cover full MTOW (33.8 N). Above that threshold, active pairs owe zero lift — every watt goes to thrust.

## SECTION 38 — H-33: VARIABLE WING AREA (18 VARIANTS)

| Variants | Configuration | Notes |
|---|---|---|
| v01-v04 | Single pair stowed (C), 3 active | W1, W2, W3, or W4. 1.2× freq boost. |
| v05 | W1+W4 stowed (C), W2+W3 active | Ends stowed, mids active. Pitch-balanced. |
| v06 | W2+W3 stowed (C), W1+W4 active | Mids stowed, ends active. |
| v07-v08 | Front or rear half stowed | W1+W2 or W3+W4 stowed. |
| v09-v11 | Stopped at neutral (B) | W1, W4, or W1+W4 stopped. Still generate lift. |
| v12-v15 | Power-boosted stow | 1.4-1.7× freq on remaining active pairs. |
| v16-v17 | Speed-adaptive stow | Auto-stow W4 above 6 m/s, W1+W4 above 9 m/s pitot. |
| v18 | Gust-adaptive auto-stow | Auto-stows W4 when sigma_roll >5°. Auto-restores. |

✓ v09-v11 (State B) are safe for TC-04. Stopped wings still generate fixed-wing lift. v01-v08 (State C) require TC-03 CT-J data before attempting.

## SECTION 39 — H-34: DYNAMIC SWEEP KINEMATICS (16 VARIANTS)

Per-stroke sweep changes within each flapping cycle:

| Stroke half | Sweep direction | Effect |
|---|---|---|
| Downstroke (0-180°) | Sweep forward | More span, stronger LEV, more lift. Pressure centre forward. |
| Upstroke (180-360°) | Sweep aft | 69% drag reduction vs extended upstroke (Tobalske JEB 2016). Weaker adverse LEV. |

Stopped-wing sweep schedule (pitot airspeed → optimal fixed-wing angle):

| Pitot airspeed | Sweep angle | Why |
|---|---|---|
| Below 4 m/s | 30° forward | Maximum span, maximum CL |
| 4–8 m/s | Transitioning neutral | Efficient cruise glide |
| Above 8 m/s | 75-85° aft | Minimum drag, minimum gust sensitivity |

✓ Variants v03/v06 (40° excursion) are gated on H-27 servo bandwidth data. H-27 measures actual achievable servo excursion within the stroke period. Conservative 20° variants (v01/v02) are safe from TC-04.

## SECTION 40 — H-35: THRUST/LIFT ROLE SEPARATION (14 VARIANTS)

Separates the aircraft into wings (stopped pairs) and engines (active pairs):

| Role | What it does |
|---|---|
| Lift wings (stopped pairs) | Held at speed-scheduled AoA. Zero motor power. Pitch servo actively trims AoA each tick. At 9 m/s: 2 stopped pairs cover full MTOW — active pairs are pure thrust. |
| Thrust wings (active pairs) | AoA at 5-8° (thrust bias, not lift bias). 1.25× frequency boost. Every watt is propulsion. |
| Stowed pairs (if any) | From H-33. Zero lift, zero drag, zero interference. |

| Variant | Config | Notes |
|---|---|---|
| v01 | W2+W3 lift, W1+W4 thrust | Middles as wings, ends as engines |
| v02 | W1+W4 lift, W2+W3 thrust | Ends as wings, middles as engines |
| v05/v06 | Single lift wing | Entry config: 1 stopped, 3 thrust — most conservative |
| v07 | Speed-adaptive | Auto: 0 lift wings below 5 m/s, 1 above 5, 2 above 8 m/s |
| v13 | Sprint: W2+W3 lift, W1 thrust, W4 stowed | Max speed attempt |
| v14 | Full composition | H-33+H-34+H-35 all active simultaneously |

✓ Start with v05 or v06 (single lift wing, 3 thrust) in TC-04. Graduate to 2-pair separation after TC-03 CT-J confirms the load budget.

## SECTION 41 — H-36: MINIMUM POWER LOITER AND FERRY (8 VARIANTS)

3 pairs stopped as fixed-wing lift wings, 1 pair at minimum frequency. Structural analogy: the MCO-1D becomes a fixed-wing aircraft with one flapping propulsion unit. Power estimate: ~15-18 W vs ~60 W normal cruise. Endurance estimate: ~3× improvement (confirmed by TC-03 data).

| Variant | Configuration |
|---|---|
| v01 | W1 active (front engine), W2+W3+W4 wings |
| v02 | W4 active (rear engine), W1+W2+W3 wings |
| v03 | W2 active, W1+W3+W4 wings — best pitch balance |
| v04 | W3 active, W1+W2+W4 wings — best pitch balance |
| v06/v06b | W3 or W2 active + frequency hunting (auto-finds minimum freq) |
| v07 | W2 active, ferry objective (maximise km/Wh) |
| v08 | W3 active, SAR loiter objective (maximise endurance × stability) |

✓ Start with v03 or v04 (W2 or W3 active) — best pitch balance. v06 frequency hunting requires stable forward flight first. Use loiter_endurance_start discovery mode.

## SECTION 42 — MORPHING SPEED DISCOVERY MODE

```
>>> morphing_speed_start
```

Runs all 48 H-33/34/35 variants in priority order. Safe variants (State B, single lift wing) first. Sprint and full-stow variants last. 45 seconds per variant. Scores: 45% speed + 35% gust + 20% efficiency. Persists leaderboard to /home/pi/knowledge/morphing_speed_registry.json.

```
>>> morphing_speed_report # show leaderboard sorted by pitot airspeed
>>> morphing_next # advance to next trial manually
>>> h33_h34_h35_disengage_all # return all pairs to normal flapping
```

## SECTION 43 — LOITER ENDURANCE DISCOVERY MODE

```
>>> loiter_endurance_start
```

Runs all 8 H-36 variants. Scores: 50% power (lower=better) + 30% altitude stability + 20% roll stability. Also scores ferry objective (km/Wh) separately.

```
>>> h36_status # power, endurance estimate, free lift report
>>> h36_loiter_report # endurance leaderboard
>>> h36_ferry_report # range/Wh leaderboard
```

# PART VIII — DISCOVERY PROGRAMME

## SECTION 44 — WHAT DISCOVERY MODE DOES

The optimizer runs continuously at 50 Hz in every flight mode. Discovery Mode gives it the whole flight to explore systematically. Four exploration layers run simultaneously:

| | |
|---|---|
| **Layer 1 — Gradient** | Small improvements to the current best configuration. |
| **Layer 2 — Biomimetic** | Tests all 33 hypotheses (180 variants) from research and novel modes. |
| **Layer 3 — Evolutionary** | Breeds new configurations from two previously good ones. |
| **Layer 4 — Random** | Escapes local optima by jumping to unexplored configurations. |

The three-axis trial validity gate ensures only accurately executed configurations are scored: AoA error <3°, sweep error <5°, phase error <10°. Trials outside these gates are discarded rather than corrupting the registry.

## SECTION 45 — RUNNING A DISCOVERY FLIGHT

| Mode | Command | Focus |
|---|---|---|
| Full discovery | disc_mode_engage | All 12 focus modes, balanced |
| Morphing speed | morphing_speed_start | H-33/34/35 priority, speed primary score |
| Loiter/ferry | loiter_endurance_start | H-36 priority, power/endurance primary score |
| Gust resilience | disc_mode_engage_focused("gust_resilience") | sigma_roll reduction |
| Hover endurance | disc_mode_engage_focused("hover_endurance") | Minimum power at hover |
| Biomimetic focus | disc_mode_engage_focused("biomimetic") | All 33 hypotheses at full rate |

```
>>> disc_mode_status # current session progress
>>> disc_mode_exit # exit and auto-export knowledge
>>> knowledge_export_usb # copy registry to USB stick
```

## SECTION 46 — ALL 33 HYPOTHESES H-01 TO H-36

| ID | Name | Description |
|---|---|---|
| H-01 | Dragonfly phase | In-phase for force, anti-phase for stability. 5v. |
| H-02 | Bat advance rotation | AoA transition before stroke reversal. Direct Re. 5v. |
| H-03 | Manta spanwise wave | Phase gradient root-to-tip. 89% prop efficiency. 3v. |
| H-04 | Pterosaur soar | 0.8 Hz, 18° AoA, needs h>30m. 1v. |
| H-05 | Hawkmoth LEV | Amplitude gradient delays LEV breakdown. 4v. |
| H-06 | Negative cascade | W4 leads W1 — reversed cascade. 1v. |
| H-07 | Split frequency | 10% inter-pair frequency differential. 3v. |

| | | |
|---|---|---|
| H-08 | Manta Strouhal | Speed-adaptive St 0.20–0.40 from pitot airspeed. 15v. |
| H-12 | Locust reversal burst | ODrive burst at stroke reversal. 3v. |
| H-13 | Plesiosaur phase | Speed-adaptive cascade phase (Muscutt 2017). 4v. |
| H-14 | Plesiosaur brake | Anti-augmentation deliberate drag. 1v. |
| H-15 | Plesiosaur dominance | Pair amplitude dominance switching. 6v. |
| H-16 | Turtle figure-eight | Dynamic sweep per stroke. Needs sweep ADC. 4v. |
| H-17 | Turtle asymmetric | Power downstroke, recovery upstroke. 1v. |
| H-18 | Turtle sweep-clap | Sweep-axis clap at downstroke bottom. 3v. |
| H-19 | Sea lion AoA | 15° thrust / 22° lift (Re-corrected). 4v. |
| H-20 | Otter advance ratio | Drag-to-lift mode switch with pitot airspeed. 5v. |
| H-21 | Sea lion fling | High-velocity clap initiation. 3v. |
| H-22 | Bat ventral clap | Upstroke ventral clap — double lift peak. 4v. |
| H-23 | Bat speed-adaptive AoA | -8° hover → +5° cruise → -3° sprint. 6v. |
| H-24 | MCO-1D AoA gradient | W1 thrust AoA, W4 lift AoA. No precedent. 4v. |
| H-25 | MCO-1D per-stroke roll | AoA differential for roll, 4× faster than amplitude. 4v. |
| H-26 | MCO-1D force decouple | Sidestep without banking. TC-02 reveals behaviour. 4v. |
| H-27 | MCO-1D servo bandwidth | Maps servo tracking at each frequency. 12v. |
| H-28 | MCO-1D asym cascade | Separate port/stbd cascade phases. 6v. |
| H-29 | MCO-1D resonance seek | Each wing finds own mechanical resonant frequency. 4v. |
| H-30 | MCO-1D power smooth | Distribute reversals to minimise bus ripple. 3v. |
| H-31 | MCO-1D Fourier path | Arbitrary 2D stroke path beyond figure-eight. 4v. |
| H-32 | MCO-1D multi-wake | W3/W4 catch multiple upstream wakes. 2v. |
| H-33 | Variable wing area | Stow (State C) and stop (State B) pairs. 18v. NEW R16. |
| H-34 | Dynamic sweep | Per-stroke ±20-40° sweep. Speed-adaptive stopped sweep. 16v. NEW R16. |
| H-35 | Thrust/lift separation | Stopped pairs=wings, active pairs=engines. 14v. NEW R16. |
| H-36 | Loiter and ferry | 3 pairs stopped, 1 at minimum freq. ~3× endurance. 8v. NEW R16. |

# PART IX — REFERENCE

## SECTION 47 — GPIO AND BUS PIN REFERENCE

| Pin | Function |
|---|---|
| GPIO 4 | DS18B20 OAT (One-Wire) |
| GPIO 5 | W3p sweep PWM |
| GPIO 6 | W3s sweep PWM |
| GPIO 7 | SPI1 CE1 — pitch ADC / HX711 SCK bench |
| GPIO 8 | SPI1 CE0 — sweep ADC |
| GPIO 9 | SPI1 MISO — both ADCs |
| GPIO 10 | SPI1 MOSI — both ADCs |
| GPIO 11 | SPI1 CLK — both ADCs |
| GPIO 12 | W3p pitch PWM |
| GPIO 13 | W4p sweep PWM |
| GPIO 16 | W3s pitch PWM |
| GPIO 17 | W1s sweep PWM |
| GPIO 18 | W1p pitch PWM |
| GPIO 19 | W4s sweep PWM |
| GPIO 20 | W4p pitch PWM |
| GPIO 21 | W4s pitch PWM |
| GPIO 22 | W2s sweep PWM |
| GPIO 23 | W1s pitch PWM |
| GPIO 24 | W2p pitch PWM |
| GPIO 25 | W2s pitch PWM / CAN adapter interrupt |
| GPIO 26 | HX711 DT — load cell data (bench only) |
| GPIO 27 | W2p sweep PWM |
| I2C1 GPIO2/3 | IMU (0x68) + barometer (0x76) |
| SPI0 CE0 | MCP2515/2518FD CAN adapter |
| UART GPIO14/15 | RC SBUS + Pixhawk MAVLink 921,600 baud |

## SECTION 48 — ODRIVE AXIS ASSIGNMENT

| Board | CAN Node | axis0 | axis1 |
|---|---|---|---|
| Board 0 | Node 0 | axis0 → W1 port | axis1 → W1 starboard |
| Board 1 | Node 1 | axis0 → W2 port | axis1 → W2 starboard |
| Board 2 | Node 2 | axis0 → W3 port | axis1 → W3 starboard |

| | | | |
|---|---|---|---|
| Board 3 | Node 3 | axis0 → W4 port | axis1 → W4 starboard |

```
CAN arb ID axis0: (node_id << 5) | cmd_id
CAN arb ID axis1: ((node_id | 0x10) << 5) | cmd_id
Wing idx → board+axis: board = wing_idx // 2, axis = wing_idx % 2
```

## SECTION 49 — SOFTWARE FILE REFERENCE

| File | Lines | Purpose |
|---|---|---|
| mco_wave_library.py | 48,001 | Main flight control. 33 hypotheses, 180 variants. 148 GS commands. |
| mco_main.py | 405 | 50 Hz flight loop. Reads all sensors. Applies wing commands. |
| mco_loadcell.py | 527 | HX711 load cell. Calibrate, tare, thrust sweep, hover detect. |
| mco_build_measurements.py | 328 | 15-measurement physical survey. Patches wave library. |
| mco_preflight_checks.py | 573 | 14 pre-flight checks. Battery interlock. Arming gate. |
| mco_rc_verify.py | 324 | RC channel mapping verification and lever interlock. |
| mco_vibration_check.py | 367 | IMU vibration sweep. SPI reliability. Tick budget. |
| mco_test_cards.py | 1,665 | TC-01 through TC-04. Automated/guided commissioning. |
| mco_log_analysis.py | 462 | Post-flight log analysis, anomaly detection, thrust curve. |
| mco_odrive_interface.py | 253 | 4-board 8-axis CAN interface. |
| mco_odrive_setup.py | 189 | Configures all 8 axes: pole_pairs, vel_limit, mode. |
| mco_ground_station.py | 660 | Laptop GS. Sensing display. Morphing/loiter reports. |
| mco_logger.py | 220 | 50 Hz CSV logger. 60+ fields per tick. |
| mco_mavlink.py | 290 | Pixhawk bridge. GPS, IMU, pitot airspeed → V_ms. |
| mco_preflight.py | 782 | Three-tier pre-flight (original). ADC and axis1 checks. |
| mco_panel.py | 485 | Touchscreen: 4 buttons, 6 margin bars. |
| mco_servo_interface.py | 110 | Servo PWM via pigpio. |
| mco_install.sh | 46 | One-step installation script. |
| mco_can_setup.sh | 36 | CAN interface setup shell script. |

## SECTION 50 — SELECTED GROUND STATION COMMANDS

| Command | Function |
|---|---|
| arm / disarm | Arm or disarm all 8 motor axes |
| emergency_stop | All amplitudes to zero immediately |
| tc01_start / tc01_confirm / tc01_status | TC-01 bench test control |
| tc02_start / tc02_status | TC-02 captive hover control |
| tc03_start / tc03_status | TC-03 truck flight control |
| tc04_start / tc04_continue / tc04_status | TC-04 free flight guidance |
| tc_results / tc_export | View all TC results, export to USB |
| servo_health_report | AoA tracking RMS all 8 wings |
| stall_detector_status | Per-wing stall likelihood |
| lift_balance_report | Port vs stbd lift estimate |
| stroke_position_status | Actual vs commanded stroke all 8 wings |
| effective_aoa_status | Effective (not geometric) AoA per wing |
| sensing_display | Full per-wing sensing panel |
| morphing_speed_start | Run H-33/34/35 speed exploration |
| morphing_speed_report | Leaderboard: best pitot airspeed per variant |
| morphing_next | Advance to next morphing trial |
| h33_status / h34_status / h35_status / h36_status | Morphing hypothesis status |
| h33_h34_h35_disengage_all | Return all pairs to normal flapping |
| loiter_endurance_start | Run H-36 loiter/ferry exploration |
| h36_loiter_report / h36_ferry_report | Endurance and range leaderboards |
| disc_mode_engage | Full Discovery Mode |
| disc_mode_engage_focused X | Focused mode (morphing_speed, biomimetic, etc.) |
| disc_mode_exit | Exit and auto-export |
| descent_ctrl_engage N | Engage N fpm descent (100=IFR standard) |
| calibrate_descent_controller | Apply TC-02 measurements to controller |
| knowledge_export_usb | Save discovery registry to USB |
| failure_briefing | Pre-flight failure performance table |
| wing_sim_request / wing_sim_approve | Wing idle simulation |

## SECTION 51 — CALIBRATION CONSTANTS PENDING TC-02

These estimates work conservatively until TC-02 provides real data. Replace with measured values immediately after TC-02.

| Constant | Current Value | Source |
|---|---|---|
| DESCENT_CTRL_HOVER_FREQ_HZ | 2.50 Hz | TC-02-03: frequency at which load cell reads zero |
| DESCENT_CTRL_K_FREQ_PER_FPM | 0.0009 | Computed by calibrate_descent_controller() from hover freq and slope |
| ACTIVE_DAMP_K_AMP | 0.40 | TC-04-05: °_amplitude per °/s roll rate in free flight |
| ACTIVE_DAMP_K_AOA | 0.15 | TC-04-05: AoA correction per °/s roll rate |
| ACTIVE_DAMP_SETTLE_TIME_S | 0.15 s | TC-04-05: time from stick release to rate <2°/s |
| AOA_CL_KP/KI/KD | 0.30/0.05/0.02 | TC-04-02/03: closed-loop AoA gains from free flight |
| GR_AOA_FAST_GAIN | 0.25 | TC-04-06: gust response fast-leg gain from turbulence data |
| H-26 behaviour | Unknown | TC-02-06: ROLL or LATERAL_FORCE classification |
| _H35_V_CRUISE_ENTRY | 5.0 m/s | TC-03: pitot airspeed at which first pair transitions to lift role |
| _H35_V_CRUISE | 8.0 m/s | TC-03: pitot airspeed at which 2-pair lift configuration activates |

## SECTION 52 — PRE-FIRST-FLIGHT CHECKLIST SUMMARY

| Done | Item | Notes |
|---|---|---|
| ✗ | mco_loadcell.py --calibrate | Calibrate with 1 kg mass — REQUIRED before TC-01 |
| ✗ | mco_build_measurements.py | Measure physical aircraft — REQUIRED before TC-01 |
| ✗ | mco_rc_verify.py | Confirm CH1-4 mapping — REQUIRED before TC-01 |
| ✗ | mco_odrive_setup.py --verify | Confirm vel_limit=150 on all 8 axes |
| ✗ | mco_preflight_checks.py --bench | All 14 checks pass — pre-TC-01 |
| ✗ | mco_vibration_check.py --quick | Baseline before first motor run |
| ✗ | TC-01 execution | 16 steps — tc01_start on ground station |
| ✗ | mco_vibration_check.py (full) | After TC-01-06, before TC-01-07 |
| ✗ | mco_log_analysis.py --tc01 | Review thrust curve after TC-01 |
| ✗ | TC-02 execution | 8 steps — tc02_start on ground station |
| ✗ | mco_log_analysis.py --tc02 | Confirm hover frequency extracted |
| ✗ | TC-03 execution (truck) | 6 steps — tc03_start on ground station |
| ✗ | CT-J curve confirmed | Required before H-33 State C / H-35 2-pair / H-36 stow |
| ✗ | TC-04 execution | 7 steps — first free flight — tc04_start |
| ✗ | Pitot airspeed verified | V_ms showing pitot airspeed (not GPS) in telemetry |
| ✗ | Float specification | Water ops hardware — post TC-02 |
| ✗ | Stroke plane controller | Post TC-03 CT-J curve |